

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. CHB-503: Organic Chemistry-III

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following:

[5 × 2 = 10]

- (a) Predict the product: Nitrobenzene $\xrightarrow{\text{Na/liq. NH}_3, \text{EtOH}}$?
- (b) Describe radical halogenation of alkanes with a suitable example.
- (c) How can cyclopropanes be prepared using carbenes?
- (d) Give a preparation method for cyclobutane from malonic acid.
- (e) What is the Wittig reaction? Give one example.

Question 2.

- (a) Explain the Benzidine rearrangement reaction. Provide evidences to support that it is an intramolecular rearrangement reaction. [7]
- (b) Discuss the conditions and mechanisms to obtain simple and modified Wittig reaction products. [7]

Question 3.

- (a) Explain the concept of a plane of symmetry and center of symmetry with suitable chiral and achiral examples. [5]
- (b) Define resolution of racemic modification and explain two chemical methods of resolution. [5]
- (c) Explain regioselectivity with a suitable example. [4]

Question 4.

- (a) What do you understand by asymmetric induction? Explain Cram's rule of asymmetric induction using a suitable carbonyl reaction. [6]
- (b) Explain prochirality with suitable examples. How can trigonal carbons be made into stereogenic centers? Explain with a reaction. [6]

(c) Write the mechanism of the Pinacol-pinacolone rearrangement.

[2]